FRANCE AND THE FIRST EUROPEAN PAPER MONEY EXPERIMENT

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'Paper money' was on the whole a pejorative term in the eighteenth century. For contemporaries, it was more or less synonymous with discredited or devalued money; nothing could preserve paper money from a 'devaluation which was inherent in its nature', so it was said.[1] Though many chose to dismiss the very idea of paper money, the recourse to fiscal experimentation by governments meant that the controversies surrounding its emission had to be fully debated. By the end of the eighteenth century, many states were financing their wars by borrowing, and in some cases by the emission of paper money. As Benjamin Franklin explained, with reference to the process operating in the rebel states during the War of American Independence, paper currency 'pays and clothes troops, and provides victuals and ammunition, and when we are obliged to issue a quantity excessive, it pays itself off by depreciation'.[2]

Yet 'the progress of . . . such systems', as Tom Paine put in 1796,[3] has not received a fully satisfactory modern treatment. James C. Riley commenced the comparative study of this experiment more than twenty years ago.[4] This article seeks to build upon his foundations and consider the extent to which the first widespread paper money experiment emerged as a response to the 'fiscal crisis' experienced by several European states in the eighteenth century.[5]

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[1] The debate on the introduction of the *assignats* is instructive in this regard. A. E Murphy, 'John Law and the assignats', *La pensée économique pendant la Révolution française*, ed. G Faccarello and P Steiner (Grenoble, 1990), p 445, F H Crouzet, *La grande inflation. la monnaie en France de Louis XVI à Napoléon* (1993), pp. 114-15 In his *Effets des assignats sur le prix du pain*, Du Pont de Nemours predicted a doubling of the price of bread and other items of popular consumption For Du Pont. J. J McLain, *The economic writings of Du Pont de Nemours* (Newark, Del , 1977)

[2] Quoted by J C Riley, *International government finance and the Amsterdam capital market, 1740-1815* (Cambridge, 1980), pp 21-2

[3] Ibid p 401. As Tom Paine put it in *The decline and fall of the English system of finance* (1796), 'gold and silver will, in the long run, revolt against depreciation and separate from the value of the paper; for the progress of all such systems appears to be, that the paper will take command in the beginning, and gold and silver in the end'

[4] Riley, *International government finance*

[5] P T Hoffman and K. Norberg (eds), *Fiscal crises, liberty, and representative government, 1450-1789* (Stanford, Calif., 1994), W M Ormrod, M. M Bonney and R. J. Bonney (eds), *Crises, revolutions and self-sustained growth essays in European fiscal history, 1130-1830* (Stamford, UK, 1999).

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Particular attention will be paid to the origins and purposes in France of the 'system' of paper money. At what point did the recognition of a 'fiscal crisis' lead to the creation of a new 'system' of deficit finance beyond merely a set of expedients, or what the economic and monetary theorist Ferdinando Galiani termed a 'currency of necessity'?[6] What follows in this article is a discussion of the first experiment in introducing paper money in France, its subsequent development in Europe and the implications for a theoretical understanding of the issues surrounding fiat money and its possible role in facilitating deficit finance and debt redemption.

I

The introduction of *billets de monnaie* in 1701 and the subsequent adoption of Law's System in France after 1716 took place against a background of both a monetary and a debt crisis [7] The revaluation of the *livre tournois* in 1726, and its relative stability until 1785, provided much needed confidence in the currency, which had previously been the sacrificial victim of an aggressive French foreign policy. under Louis XIV, the currency had been debased over forty times between 1686 and 1709. Over the longer period between 1688 and 1726 the French currency lost nearly half its value in terms of silver. There were eleven deflationary reductions in the value of the *louis d'or* by Desmaretz (controller-general of finance, 1708–15) between December 1713 and September 1715, the 'real objective' of which, it has been said, was 'to improve the fiscal situation rather than solve the monetary crisis'.[8] Alterations to the value of the specie money supply both produced segniorage gains to the crown (perhaps 250 million in December 1715 alone), and reduced the real cost of servicing the state debt. John Law particularly denounced this policy· 'raising' the money in France (that is, the policy of debasement and devaluation), he contended, 'is laying a tax on the people, which is sooner paid, and thought to be less felt than a tax laid on any other way . . . This tax falls heav[il]y on the poorer sorts of people.'[9]

Because of the substantial reminting activities in which the state was permanently involved,[10] mint bills (*billets de monnaie*) were a logical answer to provide a new credit instrument. If a delay in exchange of new coin for old

[6] In 1751 Galiani called banknotes 'mere images of money' He denounced 'that imposture, namely paper, which represents nothing actually in existence and represents nothing certain in the future' R J Bonney, 'Early modern theories of state finance', *Economic systems and state finance*, ed R. J Bonney (Oxford, 1995), p 211.

[7] M M and R. J Bonney, *Jean-Roland Malet: premier historien des finances de la monarchie française* (Comité pour l'histoire économique et financière de la France, 1993), pp 70–4, J M Félix, 'Les dettes de l'État à la mort de Louis XIV', *Comité pour l'histoire économique et financière de la France Études et documents*, 6 (1994), 603–8

[8] A E. Murphy, *John Law economic theorist and policy-maker* (Oxford, 1997), p 152

[9] Ibid

[10] F C Spooner, *The international economy and monetary movements in France, 1493–1725* (Cambridge, Mass , 1972)

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was anticipated at an individual mint, a receipt would be issued by the mint director, promising reimbursement within a short time. In October 1701, citing a general shortage of coin, the crown elected to systematize these receipts into credit instruments at an interest rate of 4 per cent.[11] The initial amounts in circulation were modest (some 6.7 million), but after another devaluation in May 1704 and a rush to exchange bills for coin following the military disaster at Blenheim in August, the repayment of bills was delayed to 1708, although rates of interest were raised to 10 per cent. The emission of further *billets* was viewed as the only way to finance the war effort, other expedients such as the *affaires extraordinaires* having dried up.[12] By the spring of 1707, there were 173 million *livres* in circulation,[13] against an estimated specie money supply of 500 million, an approximate balance of one-third paper money to two-thirds specie money.[14] The *billets* fell into rapid discount (some 55 per cent discount in cash terms by July 1709),[15] so that the public viewed them with increasing suspicion; if there was a hope for the success of paper money in this period, it rested largely on the assumption that the frequency of currency manipulation might make paper a more attractive alternative.[16] Moreover, why should interest be paid on the *billets* at all, mused Trudaine, the intendant at Lyon: if the paper was genuine money it should not carry interest ('ils ne doivent estre regardés que comme de l'argent comptant qui ne produit point d'interests par luy mesme').[17]

If it was indeed genuine paper money, then the crown had to display confidence in the expedient, and not imply that the *billets de monnaie* would shortly be revoked.[18] Yet the government showed itself to be as suspicious of paper as the general public, and issued contradictory signals, refusing at first to accept the *billets* for tax payments and later only allowing them to be used for one-quarter of the total payment in ordinary transactions, the rest having to be discharged in specie money.[19] This limited convertibility of the *billets* hastened

[11] W G Monahan, *Year of sorrows: the Great Famine of 1709 in Lyon* (Columbus, Ohio, 1993), p 43 The earlier study is A Seligmann, *La première tentative d'émission fiduciaire en France: étude sur les billets de monnaie du Trésor royal à la fin du règne de Louis XIV, 1701-1718* (1925)

[12] A[rchives] N[ationales] G^7 361, 26 Apr 1707

[13] In May 1707 there had been 173 million in circulation without interest, the declaration of 24 May 1707 fixed the limit for circulation at 72 million with interest payments to help facilitate the *cours*: AN G^7 361, 8 Nov , 1 Dec , 17 Dec 1707

[14] Murphy, *John Law*, p 116 These low levels of monetization are confirmed by local evidence, e g. at the Lyon market where there were perhaps 300,000 or 400,000 *livres* in circulation yet the great financier Samuel Bernard could export 600,000 or 700,000 *livres* at one time AN G^7 359 no 193, 17 Dec 1704.

[15] AN G^7 363, 4 July 1709 Rumours of peace, especially from abroad, increased the value of the *billets*: AN G^7 361, 5 Apr 1707

[16] AN G^7 363, 16 May 1709

[17] AN G^7 361, 13 July 1706

[18] Chamillart announced that use of them would shortly be abolished but measures would be taken against those who had made fortunes on their conversion AN G^7 362, 27 Nov 1707, ministerial comment on a letter from Trudaine

[19] Declaration of 18 Oct. 1707. AN G^7 361, 29 Oct. 1707

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the export of specie money abroad:[20] in 1715 Law still contended that the estimated specie money supply was no more than *c.* 550 million *livres* rather than the 1.1 billion other contemporaries claimed.[21] But for convertibility to take place, there had to be a secure capital fund to underwrite the emission of the *billets*: this was impossible to establish in wartime, indeed the very purpose of the expedient was to help finance the war effort.[22] In January 1709 a royal bank was to be established which would assume responsibility for the *billets*; but with a proposed capital of 12 million, the capital base of the bank was small (though it should be noted that the initial capital base of Law's bank in 1716 was half the size).[23] Within a month one of the leading shareholders had withdrawn, leading to the collapse of the bank and the prospect of a freefall in the value of the *billets*.[24] Finally, it was declared in October 1710 that from February of the following year the *billets* could no longer be used as a mechanism of payment, and provision was made to convert them into annuities (*rentes*).

II

The first experiment with paper money in France was particularly influential in the development of John Law's thinking. He considered that rather than facilitating the money supply problems during a period of foreign war, by being forcibly converted into annuities, the *billets de monnaie* had been removed from the channels of monetary circulation. Capital used in this way, he thought, 'instead of being useful to the state becomes a charge on it'. Initially, the defect of the experiment had been that the *billets* had been issued as high denomination notes, and could not be used for small transactions The French government, moreover, had been inconsistent in its thinking, using the *billets* to finance its expenditure but being reluctant to accept them in payment of tax receipts or for the purchase of annuities. Chamillart, the finance minister from

[20] If bills could be converted to cash on demand, as at Venice and Holland, the intendant at Lyon contended that it would be 'le plus grand coup d'estat que ministre puisse faire' AN G^7 360, 2 Feb 1706 A M de Boislisle (ed), *Correspondance des contrôleurs-généraux des finances avec les intendants des provinces, 1683-1715* (3 vols , 1874-9), ii 302 (no 965) *Billets* could not be used for the purchase of land or offices, a factor in the export of specie money abroad. AN G^7 361, 17 Sept 1706

[21] Murphy, *John Law*, p 153

[22] It was impossible to reimburse all holders of *billets* who might present themselves AN G^7 361, 7 Sept 1706 When the intendant at Lyon remarked on the need for an alienation of a capital sum from which interest payments for the 72 million in *billets* could be paid, Desmaretz commented. 'quels fons pourroit on aliener [?]' There was no such thing as a secure (*autentique*) alienation which could not be changed subsequently AN G^7 362, 1 and 2 Dec 1707, with Desmaretz's comments in the margin

[23] However, the capital base of Law's bank was only 6 million at the outset, of which only 375,000 *livres* was in specie Yet in two years it issued 148 5 million in banknotes. Murphy, *John Law*, pp 158-9

[24] Monahan, *Year of Sorrows*, pp 82-3 AN G^7 363, 19 Jan 1709, 24 Jan , 2 Feb. and 12 Feb 1709 'Voicy une crise de la derniere consequence si la banque ne s'establit point et que les billets de monnoye retombent en grosse perte '

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1699 until 1708, had failed to maintain specie reserves in order to ensure the convertibility of the *billets*. Above all, the government had been preoccupied with using the *billets* to raise finance for itself rather than exploiting their role as a new form of money and had forced the use of the instrument on to the public ('l'acceptation de ce credit estoit forcé, son crédit doit être volontaire').[25] In contrast, Law concluded that the range of credit instruments in Britain was wider, and the paper money supply (within his definition) was primarily issued by the private sector. In the British case, monopoly privileges for banking and trading provided the Bank of England and the East India Company with a diversified portfolio of assets, so that the profitability of these institutions was not solely a consequence of receiving interest from their holdings of government debt. Calculating the market capitalization of the Bank of England at £2.4 million and the expansion of credit at the Bank at £2 million, Law contended that in borrowing £2 million William III had produced the same effect as if the quantity of money had increased by £4.4 million.[26] Paper was preferable to specie money if its security was good ('si la seureté est bonne').[27]

Law's System (and he wanted it known as a 'System': 'here you have a chain of ideas which support one another, and display more and more the principle they flow from')[28] was thus intended to solve France's monetary and debt problems, to avoid the errors of the *billets de monnaie*, and to generate an increase in national wealth on the British model Of these problems, he was convinced that the 'huge quantity of debt with which France was overburdened was the greatest obstacle to affluence'.[29] Law had boasted that in a period of peace he would drain Britain of all its silver; given the rate of exchange in March 1720, English commentators thought that he had some prospects of success.[30] This optimistic view of the exchange rate was not to last. An investor in Law's System, Richard Cantillon, asked himself how one could reduce the interest rate to 2 per cent, expand the money supply and revalue the currency simultaneously, as Law sought to do in 1720 [31] Cantillon considered that an economic system based on an intrinsically valuable metal and its resulting price and balance of payments mechanisms functioned reasonably well. Acting as an eighteenth-century George Soros, he took

[25] A E Murphy, 'John Law's proposal for a bank of Turin (1712)', *Économies et Sociétés, Série Oeconomia*, 15 (1991), 3-29 at p 22

[26] Ibid pp 10, 19

[27] Ibid p 27.

[28] For his preference for the use of the term Murphy, *John Law*, pp 233, 320-1

[29] A. E. Murphy, 'The evolution of John Law's theories and policies, 1707-1715', *Eur Econ Rev*, 34 (1991), 1109-25 at p 1124, Murphy, *John Law*, p 234

[30] Murphy, *John Law*, pp 239-40 'The extraordinary advantage which might be made by sending gold and silver here according to the present exchange between us and France and the high price given for it at the Bank and the Mints will no doubt occasion the melting down of a great deal of our silver coin [need to lower the price of gold] since we cannot hinder the exportation of silver Mr Law may at least pay the full value for it, which he would not do at present according to the proportion in price between the gold and silver '

[31] Ibid p 231

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speculative action against the French currency, which caused Law to have him expelled from France.[32]

Initially, that is until the conversion of the General Bank into the Royal Bank in December 1718, the bank maintained nearly a 25 per cent ratio of specie reserves to banknotes; but in the course of 1719 the bank's note-issuing power was gradually increased (notes to the face value of 39.5 million were in circulation in December 1718; the figure had risen to 400 million by 25 July 1719; between February and 22 May 1720 the public's holdings of banknotes increased by 94 per cent to 2.1 billion).[33] The total emission of banknotes by the Royal Bank under Law was 2.9 billion (2,943 million *livres*).[34] However, these figures fail to give a full picture of the total liquidity within the System. To achieve this, the totals of shares plus banknotes, expressed in the value of banknotes, need to be considered, since from the end of August 1719, Law's plan was to convert the government debt into shares of the Mississippi Company. These figures reveal a growth in liquidity from 1.4 billion in August 1719 to a high of 6.1 billion in May 1720, and then a fall to 3.1 billion in July 1720 (this remained more or less the total as expressed in the value of banknotes until December 1720). By October 1720, the public held 1.33 billion in banknotes, which had a market value of only 266 million *livres*. The total liquidity of the System by December 1720 in terms of the real value of shares and banknotes together was no more than 411 million, a collapse from 3.1 billion in July. Though the writing may have been on the wall for the System from the time of the decree of 27 May, which caused a 44 per cent fall in the share price in five days, there remains some justification for Edgar Faure's dating the bankruptcy of Law's System from the public panic on 17 July 1720. On 21 July the government had responded to this panic by declaring that paper money would gradually be withdrawn from circulation by August 1721.[35]

Hoffman, Postel-Vinay and Rosenthal consider that the collapse of Law's System and the recoinage of 1726 brought in its wake 'the harshest default that France would witness until 1797', amounting to 1.5 billion *livres* Though Law's paper money did not depreciate as much as would the *assignat* in the 1790s, these three historians estimate the total loss to private lenders from both devaluation and repayments as just under a quarter of the total figure for the default, or 358 million.[36] Discussing the collapse in the monthly value of shares

[32] A. E Murphy, 'John Law and Richard Cantillon on the circular flow of income', *Eur J Hist Econ Thought*, 1 (1993), 47-62 at p 58, Murphy, *Richard Cantillon. entrepreneur and economist* (Oxford, 1986), pp 139-47, on Cantillon's exchange rate strategy

[33] Murphy, *John Law*, pp. 185, 191, 238

[34] Ibid pp 300-1, with dates of decrees authorizing the issues The table published by Neal from Harsin's figures is incomplete L. Neal, *The rise of financial capitalism. international capital markets in the Age of Reason* (Cambridge, 1990), p 69

[35] Murphy, *John Law*, pp 306-7, but noting a misprint of 1 3 million and 6.1 million in the sentence for the values of the system in August 1719 and May 1720 Ibid pp. 251, 265 for the dates of the decrees Cf E Faure, *17 juillet 1720 la banqueroute de Law* (1977)

[36] P. T Hoffman, G Postel-Vinay and J -L. Rosenthal, *Priceless markets. the political economy of credit in Paris, 1660-1870* (Chicago, 2000), pp 69, 86-7

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and banknotes in the System after July 1720, Antoin Murphy comments: 'these declines were to prove disastrous for those who had held on to their shares and banknotes. However, the Crown greatly benefited from these developments in that its debt was reduced to a fraction of its former level. Though Law acquired the reputation of leaving France bankrupt this conclusion has to be revised in the light of the considerable improvement in the state's financial situation. Law's System certainly bankrupted the creditors of the state; the state, however, as a debtor was a net gainer.'[37] It was the contradiction between the contradictory demands of money creation and debt management which, in Law's view at least, had led to the collapse of his System: 'he sacrificed the reputation he had acquired, through the establishment of his bank . . to the extreme desire which he had to re-establish the state's affairs by the extinction of all the debt'.[38]

III

It is now appropriate to delineate an approximate chronology and scale of activity for the European and colonial experience of paper money. Denmark was the next state to introduce a paper currency, doing so to supplement tax revenues during the Great Northern War. The French *billets de monnaie* 'probably served as a model' for the Danish paper currency introduced in 1713.[39] The first issue in 1713 was of 400,000 *rigsdalers*, and the maximum in circulation was one million (1716–20); but all these notes had been withdrawn by 1728.[40] Friis and Glamann comment: 'certain payments, *inter alia* customs duties and taxes, were to be made half in coins, half in notes. The paper money was never accepted in general circulation as being equivalent to coins.' The paper was 'to some extent convertible into 5 per cent bonds issued by the Treasury'. Issues recommenced in 1737 (233,000 *rigsdalers*), and expanded rapidly in the 1760s until the end of the War of American Independence (the high point was over 6 million in circulation in 1764). After ministerial changes in 1784, the issues of paper were reduced somewhat (to 1.36 million), but by 1789 banknotes had regained the level of 1783 (1.56 million). By 1800 there were 1.93 million *rigsdalers* in circulation.[41]

In Sweden, domestic tax revenues expanded less rapidly than note supply and the value of the *riksdaler* in exchange transactions fell accordingly after 1741. 2 1 million *riksdalers* were issued in that year, but the amount of notes in circulation

[37] Murphy, *John Law*, p 308 Though in nominal terms, the debt may have somewhat increased R. J. Bonney, 'France, 1494-1815', *The rise of the fiscal state in Europe, c. 1200-1815*, ed R. J Bonney (Oxford, 1999), p 147

[38] Murphy, 'John Law and the assignats', p 441; Murphy, *John Law*, p 122

[39] A Friis and K. Glamann, *A history of prices and wages in Denmark, 1660-1800 I* (Copenhagen and London, 1958), p 7

[40] Ibid. p 10 (table 3)

[41] Ibid pp 11-13 (tables 5 to 7), Riley, *International Government Finance*, pp 136–41, R. J Bonney, 'Revenues', *Economic systems and state finance*, ed R. J. Bonney (Oxford-ESF, 1995), p 471

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remained below 5 million until 1756. The first great expansion occurred during the Seven Years War, when the amount of notes in circulation rose from 13.8 million *riksdalers* in 1755 to 44 million in 1763; there was a reduction to 31.8 million by 1769. Current account deficits during the Russian war were met by foreign loans and further emissions of paper, especially *riksdaler riksgälds*, which were not convertible into silver: Sweden had passed from the silver standard reintroduced in 1775 to a paper standard by 1789-90. Notes remaining in circulation fell below 8 million after 1776 but rose above 10 million in the years 1789-1803, with a high point in 1801.[42] Until the currency conversion of 1803, Sweden had two parallel standards: these comprised Riksbanknotes or *riksdaler banko* (which were convertible into silver), and National Debt Office bonds or *riksdaler riksgälds* (which were not convertible into silver)

In the Austrian lands, there were warnings in the late 1760s and early 1770s of the danger of creating an unredeemable paper money which would sink to a discount as had the French *billets d'État*. These predictions proved correct in the two decades of inflation of paper money after 1787.[43] Paper money (*Bankozettel*, the term used until 1811) had been issued before 1761 and reached 18 million florins in circulation by 1763 and 28 million by 1790 (at which date the public debt stood at 400 million).[44] Some 13 million florins in paper were redeemed between 1785 and 1788. During the Napoleonic wars, the amounts in circulation continued to rise exponentially and so too did the discount rate. From 46.8 million florins in circulation in 1796, by 1806 they had reached 450 million with a discount rate of 175; by 1809 they had reached 846 million with a discount rate of 469, in March 1811, the total in circulation was 1,061 million, with a discount rate of 883.[45] The paper was subsequently renamed 'redemption bonds' (*Einlösungsscheine*), issued at the rate of one new for five old, of the same face value: in February 1812, the volume of 1,061 million *Bankozettel* was reduced to 212.2 million in redemption bonds. Subsequently bonds repayable on the land tax in twelve years time known as *Antizipationsscheine* were issued after 1813. By 1815 there were still 635 million florins of paper in circulation, although the discount rate had improved to 408.[46] In June 1816, when the total of paper in circulation had reached 679 million, there was a second devaluation at the rate of one new for two-and-a-half old, of the same face value. This failed to halt a new discounting of the

[42] L. Jörberg, *A history of prices in Sweden, 1732-1914* (2 vols , Lund, 1972), i, 78-85, ii, 127-44, Riley, *International government finance*, pp. 144-51 Source subset Swedish money supply, 1741-1803 ESFDB /rjb/sweden/swed003

[43] P G M. Dickson, *Finance and government under Maria Theresa, 1740-1780* (2 vols , Oxford, 1987), ii 72-3

[44] Ibid ii. 115, 132, 134-8, 155-6, R J Bonney, 'The eighteenth century II The struggle for great power status and the end of the old fiscal regime', *Economic systems and state finance*, ed. Bonney, p 362

[45] R. Horvàth, 'Monetary inflation in Hungary during the Napoleonic wars', *J Eur Econ Hist*, 5 (1976), 651-62 at p 656

[46] C A. Macartney, *The Habsburg Empire, 1790-1918* (1968), pp 179, 183, 194-8, Horvàth, 'Monetary inflation in Hungary during the Napoleonic wars'

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paper money, reaching a maximum discount of 378 in January 1817. At the time of a report of the Commissions of Redemptions and Amortizations in April 1822, 450 million in paper still remained to be extinguished; 426 million had been converted by 1834, but it was not until the 1840s that the last vestiges of the paper currency expanded so massively in the Napoleonic era were eliminated.[47] As a perspective on the scale of the paper money inflation, it should be noted that the estimate for the total volume of currency in the Habsburg Empire at the outbreak of war with France in 1792 was less than 240 million florins. In nominal terms, the volume of paper had expanded this money supply almost five times before the first devaluation of 1812. By 1820, the total depreciation of the paper florin amounted to about 92 per cent, a considerably greater fall than was to occur in Russia in the same period.[48]

Paper money was first considered in Russia in 1762. An *assignat* bank and paper money had been established at the end of 1768 to help finance the First Russo-Turkish War (1768-74). 2.6 million in *assignat rubles* were issued in 1769; by 1771 the circulation had reached 10.6 million, rising to 20 million in 1774 (when Catherine II announced that this would henceforth be the ceiling on issues)[49] and 40 million in 1784. The conversion of *assignat rubles* into silver was suspended by the government in 1786,[50] though a new ceiling of 100 million, 'never' to be exceeded, was imposed.[51] During the Second Russo-Turkish war (1787-92), the Assignat Bank became the credit agency of the Treasury. Gradually the market value of the paper against silver began to plummet. There were 100 million *assignat rubles* in circulation in 1787, with a market rate of 97.1; 210 million in 1799, with a record low of 67.5. By 1808, there were 477 million in circulation, with a market rate of 53.5. Michael Speransky, the finance minister in Russia, argued in his reform plan of early 1810 that the emission of *assignats* was a form of indeterminate taxation, though imposed permanently; they were 'papers based on suppositions. Not having any intrinsic value, they are nothing else but hidden debts.' 'The qualities of all notes of credit', he concluded, 'are proportional to the capital on which they are based.' Paper money based on a standard (for Speransky, it was silver) had to be issued in proportion to the availability of silver bullion, otherwise speculation would destroy its basic function.[52] By the time that Speransky produced his financial report in 1810 there were 579.4 million *assignat rubles* in circulation and the market rate had declined by 69.1 per cent. Speransky had hoped to limit the emission to 577 million; but the budgetary deficit of 1812-14, the culmination of the war against Napoleon,

[47] Horvàth, 'Monetary inflation in Hungary during the Napoleonic wars', p 659

[48] W M. Pintner, *Russian economic policy under Nicholas I* (Ithaca, N Y , 1967), p 189

[49] J P LeDonne, *Ruling Russia. politics and administration in the age of absolutism, 1762-1796* (Princeton, N J , 1984), p. 260

[50] A Kahan, *The plow, the hammer and the knout: an economic history of eighteenth-century Russia*, ed R. Hellie (Chicago and London, 1985), p. 314

[51] LeDonne, *Ruling Russia*, p 263

[52] M Raeff, *Michael Speransky statesman of imperial Russia, 1772-1839*, rev edn (The Hague, 1969), pp. 88-9

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required further issues of paper (over half the deficit of 360 million was covered by the issue of 191 million in *assignats*). By the end of the Napoleonic wars in 1815 there were 825 million in circulation, with a market rate of 23.8.[53] *Assignats* continued to be printed until 1817, by which time the amount in circulation reached 836 million (some 254.6 million had been issued since 1812). After this date, they began to be redeemed at a fluctuating rate of exchange of about four *assignat rubles* to one silver *ruble*. As a result, the proportion of *assignats* within the total national debt began to fall. In 1796, *assignats* had constituted 76 5 per cent of the national debt (156.7 million out of a total of 205.2 million calculated in *assignat rubles*); in 1825 the proportion had fallen to 45.1 per cent (595 million out of a total of 1.3 billion [1,323 million] calculated in *assignat rubles*).[54]

In Spain, 452 million in paper money (*vales reales*) were issued during the war with Britain (1779-83), with a depreciation of 13 per cent A further 99 million were issued by 1789 The total deficit for the four years 1793-6 was estimated at 1,269 million *reales* of which about three-quarters (964 million) was covered by paper issues (*vales reales*). A further almost 800 million were issued in April 1799, with the value falling to between 30 and 45 per cent below par. Of the 2 3 billion [2,315 million] *vales* placed in circulation after 1780, 121 million were redeemed before 1800 and 300 million between 1800 and 1804 [55] In 1805, there remained 1,896 million in circulation,[56] a figure which had scarcely changed by May 1808 (when the figure of 1,893 million was reported to Napoleon).[57] While the expansion of the *vales* in circulation was a consequence of the fiscal crisis of Carlos IV's reign, it was by no means the total debt. Approximately 1,353 million of the 1,653 million from sales of disentailed church lands and redeemed *censos* constituted an additional debt of the Bourbon monarchy.[58]

It was above all in Revolutionary France that paper money was introduced with a vengeance. Interestingly, in the debate over the introduction of the *assignats* in the National Assembly, contemporaries drew the distinction between the church lands, which provided the notional collateral on which the *assignats* were assigned, and the notes issued by Law, the collateral for which was provided by the false of hopes of discovering precious metals in

[53] Riley, *International government finance*, p 156, J De Bloch, *Les finances de la Russie au xixe siècle. historique et statistique* (2 vols , 1899), P A Khromov, *Ekonomicheskoe razvitie Rossii, 1800-1917* (Moscow, 1950) Source subset Russian money supply, 1769-1815. ESDB / russia/rusd005

[54] J P LeDonne, *Absolutism and ruling class· the formation of the Russian political order, 1700-1825* (New York and Oxford, 1991), pp 282-3 For Russia. R Hellie, 'Russia, 1200-1815', *The rise of the fiscal state*, ed Bonney, ch 15

[55] R Herr, *Rural change and royal finances in Spain at the end of the Old Regime* (Berkeley, Calif , 1989), p 151

[56] M Artola, *La hacienda del antiguo régimen* (Madrid, 1982), p 451

[57] R Herr, 'El experimento de los vales reales, 1780-1808', *Dinero y crédito, siglos xvi al xix*, ed. A Otazu (Madrid, 1978), p 124

[58] Herr, *Rural change and royal finances*, pp 122, 151 For Spain J Gelabert, 'Castile, 1504-1808', *The rise of the fiscal state*, ed Bonney, ch 6

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Louisiana; Law's banknotes had been 'mortgaged on a chimera', or worse still, on ministerial lies.[59] Much of the early confidence in the *assignat* scheme in France after 1790 originated in its land security, that is to say, in the belief that the paper money would be redeemed through the sales of *biens nationaux*, which were estimated at a value in excess of 2 billion *livres*.[60] (Necker, for example, provided an estimate of 2,756 million.)[61] But leaving aside the difficulties in timing and the perilous nature of the operation, the actual sales of church lands were disappointing, amounting to not much more than 900 million in *assignats* at stable values.[62] Necker had envisaged the possibility that, at least in public opinion, the emission of paper money might exceed the value placed on the church lands; it would therefore become inevitable that people would become unwilling to retain *assignats* in the long term, while prices would double or treble.[63] Moreover, by any reckoning the proposed first emission of 1.2 billion *assignats* (a total in circulation which was not achieved in the end until 1791) would have had a very considerable impact on the money supply in France, which was thought to be in the region of 2 billion Pledging all the church lands by issuing *assignats* to the presumed value of the church lands (2 billion) would, in nominal values, have doubled the money supply.[64]

Though taxation provided the French Revolutionary government with 13 per cent of all revenues between 1790 and 1795, the paper *assignats* were deliberately expanded both to cover the deficit and to remedy the shortfall in revenue. They provided 7,000 million *livres* in revenue (at stable *assignat* values of 1790), equivalent to fourteen years of normal revenue. The gulf between revenue and expenditure became a yawning chasm after the declaration of war in April 1792, while after January 1793, the issue of new *assignats* became the crucial measure for financing the war. The accounts of September 1793 dealt with a twenty-six-month period from 1 July 1791, of which sixteen months had seen hostilities. 5,600 million (or 58 per cent) of the 6,800 million in revenues which had financed the war effort arose from the creation of *assignats*; tax revenues and the income from national lands were much lower, amounting respectively to 368 million and 130 million.[65]

Inflation reached 80 per cent a month in June 1795, while by December of that year, when the government had issued 29.2 billion *assignats*, it decreed a

[59] Murphy, 'John Law and the assignats', pp 441-2.

[60] E. N White, 'The French Revolution and the politics of government finance, 1770-1815', *J Econ Hist*, 55 (1995), 239.

[61] J Necker, *Sur l'administration de M Necker par lui-même* (1791), p 152

[62] The nominal figure was 2,100 million S E. Harris, *The assignats* (Cambridge, Mass., 1930, repr New York, 1969), p 85 The law of 18 March 1796 noted 'la disproportion entre la quantité en émission et la valeur du gage' quoted in Crouzet, *La grande inflation*, p 412 Necker had recognized the need to establish a close proximity between the value of church lands and the amount of paper in circulation Necker, *Sur l'administration de M Necker*, pp 133-4

[63] Necker, *Sur l'administration de M Necker*, p 132

[64] Murphy, 'John Law and the assignats', p 443

[65] Belhoste in *État, finances et économie pendant la Révolution française. Colloque tenu à Bercy les 12, 13, 14 octobre 1989 à l'occasion du Bicentenaire de la Révolution française* (Comité pour l'histoire économique et financière de la France, Paris, 1991), pp 323-4

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limit of 40 billion on the total issue This ceiling had been reached by 19 February 1796, when all the printing plates and equipment were solemnly destroyed. At this moment, 41 billion had been issued, but only 35.2 billion remained in circulation.[66] In 1788, the king could have balanced the budget by repudiating the foreign debt completely and reducing interest rates on the domestic debt to 6 per cent. Nine years later, in 1797, the state defaulted on two-thirds of its debt, a loss of 2.6 billion *livres* to the government's creditors.[67] Between 1789 and 1795, inflation wiped out over 99 per cent of the currency's value, costing lenders in the private market 1.67 billion *livres*, as the stock of private debt dropped to nearly zero. Compared to these losses, Hoffman, Postel-Vinay and Rosenthal contend that 'the revenue raised by printing money was small. The revenue - the inflation tax - came to less than half a *livre* for every *livre* that was redistributed . . In order to raise some two billion 1789 *livres* by printing *assignats*, the revolutionary governments provoked losses of nearly equal magnitude within the private credit market.'[68]

IV

The experiments in fiat money analysed above demonstrate a general characteristic: all, in broad terms, failed. They may have helped manage a deficit, but they did so at a tremendous cost in terms of inflation and a deteriorating exchange rate, with resulting damage to the 'real' domestic economy. In 1802, Henry Thornton contrasted the British experience, where the Bank of England was 'quite independent of the executive government', with that of the continent, where the national banks were 'in the most direct and strict sense government banks' Government loans were harder to secure on the continent than in Britain, he contended:[69]

> Those governments, therefore, have in times even of moderate difficulty, no other resource than that of extending the issue of the paper of their own banks; which extension of issue naturally produces a nearly correspondent depreciation of the value of the notes, and a fall in the exchange with other countries, if computed at the paper price. The notes, moreover, being once thus depreciated, the government, even supposing its embarrassments to cease, is seldom disposed to bring them back to their former limits . . . The expectation of the people on the continent, therefore, generally is, that the paper, which is falling in value, will, in better times only cease

[66] Crouzet, *La grande inflation*, table B, p. 572 , White, 'The French Revolution and the politics of government finance', pp 246-7, E S Brezis and F H Crouzet, 'The role of *assignats* during the French Revolution: an evil or a rescuer?', *J Eur Econ Hist*, 24 (1995), 7-40

[67] Hoffman, Postel-Vinay and Rosenthal, *Priceless markets*, pp 197, 200

[68] Ibid pp. 201, 206

[69] H Thornton, *An enquiry into the nature and effects of the paper credit of Great Britain (1802)* . ., ed F A. v Hayek (1939), pp 107-8

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> to fall, or if it rises, will experience only an immaterial rise, and this expectation serves of course to accelerate its fall.

Thornton went on to observe that the depreciation of the continental banknotes had been 'apt to originate . . . in the state of the exchanges'. He continued:[70]

> The unfavourable exchange has produced a difference between the value of bullion and that both of the current paper of those banks and of the current coin; and, when this difference has become permanent and considerable, a discount on the paper has established itself; in other words, coin has ceased to bear the price of paper, and has taken the price of bullion, and from that time the paper alone has passed at the reduced rate.

Significantly, Thornton cited the French decrees issued during the period of Law's System as the example of an excessive emission of paper leading to an unfavourable exchange rate [71] In contrast to the experiments described above, the policies of the Directors of the Bank of England were a (relative) model of fiscal rectitude. The Bullion Report of 1810 noted the rise in the annual average of banknotes in circulation from £13.3 million in 1798 to £19 million in 1809 yet considered it 'vain' to think up clear rules for 'the exact exercise' of a discretionary issue of paper money.[72] As Michael D. Bordo and Eugene White contend, 'money creation did not make a large contribution to war finance' in Britain, though it 'did give the government critical flexibility in short-term finance and debt management'.[73]

In spite of Thornton's advocacy of the British 'paper pound', the early nineteenth century witnessed a change in attitudes, which was for the most part a triumph of monetary orthodoxy. David Ricardo, self-interested though he may have been in his criticisms of the Bank of England,[74] managed to convince public opinion in Britain that 'all the evils in our currency were owing to the over-issues of the Bank, to the dangerous power with which it was entrusted of

[70] Ibid p 249

[71] Ibid. pp. 250-2 Murphy, *John Law*, pp 379-80, provides a more comprehensive list of decrees and edicts during Law's System

[72] E Cannan, *The paper pound of 1797-1821 a reprint of the Bullion Report with an Introduction* , 2nd edn (1925), pp xliv-xlv, 53, 55 For Larry Neal, 'the quarter century of a floating pound sterling from 1797 to 1821 helps us to discern an economically logical link between the French Revolution and the British industrial revolution' Neal, *The rise of financial capitalism*, p 222. However, this conclusion relates chiefly to the external exchange rate aspect of the 'paper pound' rather than the (potentially inflationary) internal value of the paper on which most contemporary commentators focused

[73] M D Bordo and E. N White, 'A tale of two currencies British and French finance during the Napoleonic wars', *J Econ Hist*, 51 (1991), 303-16 at p 311, idem, 'British and French finance during the Napoleonic wars', *Monetary regimes in transition*, ed M D Bordo and F Capie (Cambridge, 1994), p 256, cite Mitchell and Deane's figures for Bank of England notes during the war years from 1793 to 1815

[74] Ricardo's hostility to the Bank of England may have been explained by his stock exchange interests N J Silberling, 'Financial and monetary policy of Great Britain during the Napoleonic wars', *Q J Econ*, 38 (1924), 214-33, 397-439, at pp 427-9

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diminishing at its will, the value of every monied man's property . . .'[75] Ricardo quoted Adam Smith's denunciation of debasement as a 'juggling trick . . . so easily seen through, and at the same time so extremely pernicious'.[76] Ricardo claimed that 'these observations . . . are equally applicable to a depreciated paper currency',[77] questioned the independence of the Bank of England,[78] and specifically rejected an interpretation of Adam Smith's words on paper money that might justify what he regarded as the excessive number of banknotes in circulation in England in 1810.[79] (Adam Smith's discussion of paper money in the *Wealth of Nations* had given limited approval to Law's 'splendid but visionary ideas'.)[80]

V

The French experience in the early development of fiat money suggests several contrasting conclusions. The first possible conclusion is a purely negative one. Law's System failed, and it failed because it was a theoretical impossibility. Law believed that an insufficiency of money hampered economic growth. Metallic or specie money was unreliable and the system needed to be replaced by what Murphy terms 'a more accommodating paper cum credit money system'. For David Ricardo, writing in the first decade of the nineteenth century, an approach such as that of Law was open to a number of objections. He denied it could increase wealth ('I cannot agree that any addition to the money of a country produces riches, and population'), while it would be discriminatory, advantaging some and disadvantaging others, or alternatively operating as a tax on one part of the population for the benefit of another. The exportation of specie money would be checked only by a reduction in the paper money supply, which 'if carried sufficiently far' would 'produce importation'.[81]

A second conclusion stresses the modernity of Law's conceptualization of his System, and the possibilities for its success in the early eighteenth century. Antoin Murphy argues that[82]

> this movement from a specie-based outside money system to a dual product inside money system represented a revolutionary step for economic man, a step that helped smooth the establishment and development of capitalism . . . Specie money facilitates the exchange

[75] [D Ricardo,] *The works and correspondence of David Ricardo*, ed P Sraffa and M H Dobb (10 vols., Cambridge, 1951-5), iii. 21

[76] A Smith, *An inquiry into the nature and causes of the wealth of nations*, ed. R. H. Campbell, A S Skinner and W. B Todd (2 vols , Oxford, 1976), ii. 930

[77] *The works and correspondence of David Ricardo*, ed Sraffa and Dobb, iii 97.

[78] Ibid iii 98

[79] Ibid. iii. 236–7.

[80] Smith, *An inquiry into the nature and causes of the wealth of nations*, ed. Campbell, Skinner and Todd, i 317 (but there would seem to be inconsistencies in Smith's position on paper money: ibid i 296, 300); Bonney, 'Early modern theories of state finance', *Economic systems*, ed Bonney, p 220

[81] *The works and correspondence of David Ricardo*, ed Sraffa and Dobb, iii 317, 325, 334, 393.

[82] Murphy, *John Law*, pp. 106–7.

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> of commodities; bank money has the same function, but it also has another attribute which specie money does not possess, namely that of enabling borrowers to utilize money that is held by the bank . . . [E]ven if the depositor is not involved in the expenditure decision, a borrower is able to utilize the funds which the depositors have lodged with the bank

Nevertheless, though an advocate of the modernity of Law's ideas, Murphy accepts that in managerial terms he faced considerable difficulties. The great danger of Law's scheme was an excessive expansion of the money supply, particularly through the monetization of the shares, an 'overheating' of the system leading to the export of specie money abroad. One contemporary estimate suggests that by the end of November 1719 Law had expanded circulation by 5.2 billion (640 million from the issue of banknotes and 4.56 billion from the increase in the value of shares). Yet by Law's own estimate, France could maintain a monetary circulation of no more than 3 billion in banknotes and shares (on the basis that Britain had a circulation of 1 billion and that France was capable of having a trade three times greater) A decree of 1 December 1719, which allowed creditors to demand of their debtors to be paid in banknotes rather than specie, was the first attempt by Law to force the public to use paper money rather than specie and thus reduce excess liquidity in the System.[83] This was the beginning of a series of mandatory policies to try to force specie money out of circulation in the early months of 1720.[84] Law recognized that the System had overheated, and in the measures proposed by the decree of 21 May tried to take steps to stabilize it, withdrawing 2 billion *livres* from circulation by reducing banknotes and shares by 44 per cent from 4.5 billion to 2.5 billion. Law was not allowed to implement these measures, which were revoked on 27 May. On 21 July, it was finally recognized that the Bank would be wound up: by August 1721 no banknotes would remain in circulation.[85] Law was no longer in control of events after 27 May: the collapse of the System is explicable by his political weakness and isolation and by the fact that, in managerial terms, it was difficult to implement. Law wanted to deflate the System and bring it more into line with the real economy, but the Regent hesitated too long before doing so and then took fright at the hostility of public opinion to the measures enacted.[86] There is much to be said for the argument that the potential beneficiaries of any system of paper money might change, and that the 'rules of the game' for the emission and redemption of fiat money were an essential aspect of the success of any scheme.[87]

[83] Ibid. pp 210-11
[84] Ibid. pp. 220-1
[85] Ibid. pp 245, 265, 272
[86] Ibid. pp 248, 254
[87] The author is grateful to Professor Rafael Torres Sánchez for this comment Professor Torres takes the example of issue of paper money in smaller denominations, which was popular with the merchants in Spain, but not with the large-scale export merchants Another example which one might add is that of the competing interests of creditors and gentry in Russia *c* 1840 at the time of the conversion of *assignats* into silver The concern for fiscal stability in the end proved more important than the interests of the gentry debtors Pintner, *Russian economic policy under Nicholas I*, p 208

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A third possible conclusion argues the case that a scheme such as Law's was theoretically and managerially possible in the early eighteenth century but only under certain constitutional circumstances. The *locus classicus* of this viewpoint was the remark of the great financier Samuel Bernard predicting the rejection of Law's bank proposal by the king in the last months of Louis XIV's reign, 'there being no foundation', he is reported to have said, 'for the bank he proposes in a country where everything depends on the king's pleasure'.[88] This viewpoint was categorically rejected by Law ('Le gouvernement arbitraire [*sic*: absolu] de France n'a pas été la cause que les billets de monnoye ont manqué'). Instead, he emphasized technical distinctions between the British and the French systems in the reign of Louis XIV, which meant that the French credit structure had ruined commerce, while the British had succeeded. 'The establishment of the Bank [of England had] restored order to the [English] finances and abundance in trade and supported the Crown and the state during two long wars which cost proportionately larger sums than those experienced by France.'[89] The process of revenue integration and the increasing tendency towards authoritarianism under Law's System were harbingers of profound changes indeed. It was, however, a matter of particular significance that the *Parlement* of Paris could not be brought voluntarily to co-operate with Law's System. On 26 August 1718, the *Parlement* of Paris was prohibited from interfering in the administration of finance or in any business concerning the government of the state. The great point of dissension had been Law's proposal 'to make all payments through the bank and to multiply money to infinity through credit. This proposal was resisted by everyone.'[90] The *Parlement* of Paris continued to play a spoiling role with regard to tax increases in the later eighteenth century; and, after 1789, the spoiling role was taken over by the National Assembly.[91]

The contrast with the British experience could not have been greater. Barry R. Weingast has recently restated the British success in managing sovereign debt (a rise from £6.1 million in 1694 to £16.7 million in 1697 in the first four years of the Bank of England's existence) in the context of the dramatically

[88] Remark of Samuel Bernard to the English ambassador (25 July 1715), quoted by C.-F. Lévy, *Capitalistes et pouvoir au siècle des lumières* (3 vols., The Hague, 1969–80), ii. 135–6, 155 n. 108.

[89] Murphy, 'John Law's proposal for a bank of Turin (1712)', pp. 20–1, 24. Though Law conceded (p. 21) that 'cette raison [i.e. the argument that 'le peuple n'a pas tant de confiance dans un gouvernement absolu, que dans une monarchie civile comme celle d'Angleterre'] auroit pu contribuer a faire manquer le credit en France' he believed that credit had not been greatly used in France and that the ministers (chiefly Chamillart) had not established it on correct principles: Murphy, *John Law*, p. 142.

[90] Murphy, *John Law*, pp. 177, 182, T. E. Kaiser, 'Money, despotism and public opinion in early eighteenth-century France: John Law and the debate on royal credit', *J. Mod. Hist.*, 63 (1991), 1–28, R. J. Bonney, 'The state and its revenues in *ancien régime* France', *Hist. Research*, 65 (1992), 150–76 at pp. 171–3. Law's views are placed in context by Bonney, 'Early modern theories of state finance', in *Economic systems and state finance*, ed. Bonney, pp. 206–8.

[91] R. J. Bonney, 'What's new about the new French Fiscal History?', *J. Mod. Hist.*, 70 (1998), 639–67, idem, 'The struggle for great power status', p. 348, idem, 'France, 1494–1815', p. 162.

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increased penalties for default following the institutional changes of the Glorious Revolution. Defaulting on a loan required the co-operation of Parliament. 'Representative institutions therefore proved central to the fiscal and military success of England following the Glorious Revolution ' The centrality of Parliament to the process of tax agreement and enforcement helped to ensure a more rapid rise in taxation than occurred in France during the eighteenth century.[92] Much larger amounts of public credit could be raised more cheaply than in France because of the lower 'default risk'.[93] By 1790, on the eve of Britain's twenty-two year military struggle with France, it held a debt of £244 million, nearly fifteen times the annual revenue for that year.[94] Yet the banknote circulation rose from only £4.3 million in 1750 to £10.7 million in 1790. The face value of the national debt continued to rise in the war years against France, from £290 million on the eve of the war to £862 million by 1815. The increased borrowing of the British government resulted in interest payments to creditors totalling some £236 million between 1793 and 1815, almost the value of the debt itself in 1790. Yet the increase in the circulation of banknotes in Britain remained modest. The sixth-monthly average of banknotes in circulation rose from a low point of £9.2 million in the second half of 1796, before the Suspension Act of February 1797, to a peak of £29.5 million in the second half of 1817, an expansion of more than three times the money supply in nominal terms.[95]

A fourth possible conclusion is that a development such as Law's System was both theoretically and managerially possible at the time, but only under specific circumstances of, or developments in, market understanding. The first and most important of these, emphasized by Law in his criticisms of the *billets de monnaie*,[96] was that the circulation of fiat money should not literally be by fiat ('let it be done'), in the sense that it should not be forced on the public. Though vigorously repudiated by Ricardo, this was in essence the position of

[92] P Mathias and P K. O'Brien, 'Taxation in England and France, 1715-1810', *J Eur Econ Hist*, 5 (1976), 601-50, J C. Riley, 'French finances, 1727-1768', *J Mod Hist*, 59 (1987), 209-43, W Fritschy, 'Taxation in Britain, France, and the Netherlands in the eighteenth century', *Econ Soc Hist Netherlands*, 2 (1990), 58-79 The theme is generalized in Hoffman and Norberg, *Fiscal crises, liberty, and representative government, 1450-1789* Also P T Hoffman and J.-L. Rosenthal, 'The political economy of warfare and taxation in early modern Europe historical lessons for economic development', *The frontiers of the new institutional economics*, ed J N Drobak and J V C Nye (San Diego and London, 1997), pp. 31-55 at p 53 'The portion of the economy that suffered most [under absolutism] was the financial market, which was almost always under royal control.'

[93] F R. Velde and D. R. Weir, 'The financial market and government debt policy in France, 1746-1793', *J Econ Hist*, 52 (1992), 1-39.

[94] B R. Weingast, 'The political foundations of limited government. Parliament and sovereign debt in 17th- and 18th-century England', *The frontiers of the new institutional economics*, ed. Drobak and Nye, pp 213-46 at pp. 234-7.

[95] B R. Mitchell, *European historical statistics, 1750-1975*, 2nd revised edn (Basingstoke, 1981), p. 706, Bonney, 'The struggle for great power status', pp 380-1, P. K. O'Brien, 'The impact of the Revolutionary and Napoleonic wars, 1793-1815, on the long-run growth of the British economy', *Review Fernand Braudel Centre*, 12 (1989), 335-95, idem, 'Public finance in the wars with France, 1793-1815', *Britain and the French Revolution, 1789-1815*, ed. H. T Dickinson (Basingstoke, 1989), pp 165-87.

[96] Murphy, *John Law*, pp 117-20.

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the Directors of the Bank of England during the bullion controversy in 1810: 'a paper money cannot be excessive, if no one is obliged to take it against his will...'[97] Yet Law himself was not consistent in his commitment to voluntarism. In the endeavour to establish a specie-less France in 1720, Law was prepared to allow for coercion. According to the declaration of 11 March 1720, silver was to be officially demonetized on 1 January 1721; a further decree of 19 March 1720 banned the import of gold and silver specie and bullion into France to prevent speculative conversions between specie and paper [98]

All the European experiments in the expansion of paper money supply in the period before *c.* 1820 had started as small, voluntary, schemes which, under the pressure of events, had been converted into a forced system. To the extent that the governments such as those of Spain, Austria and Russia expanded their paper supply mainly in wartime, when they were acting under duress because of the wartime emergency, the expectation of the public was that a return to peace would bring about a regularization of the system. 'Fiat money may be issued to cope with the emergency', write Michael D. Bordo and Forrest Capie, 'but if the public's perception is that this is temporary then a different outcome will emerge from that which will emerge when the public does not trust the government to restore the status quo.'[99] Bordo and Eugene White argue that this precondition of trust existed in Britain during the era of the inconvertible paper pound between 1797 and 1821,[100] but it was clearly not true of France, Spain, Austria or Russia during the period of the Revolutionary and Napoleonic wars. And as for Law's System, it had not even been adopted in a period of foreign war. In this respect, the experiment of the *billets de monnaie* under Chamillart and Desmaretz had an apparent advantage. Bordo and Capie ask whether a period of hyperinflation can be viewed as a monetary regime.[101] Clearly periods of rapid inflation are ones in which market information cannot be aggregated sufficiently fast to provide potential investors in government stocks with the information they require.[102]

[97] *The works and correspondence of David Ricardo*, ed Sraffa and Dobb, iii. 362 'This seems to be the source of all the errors of these practical men it is of little importance whether the state forces a paper circulation, or whether it be issued by a company only when demanded by the public, in discounting good bills, the effects of an excessive issue will be the same ' Ibid iii 338 'The value of bank notes [is] not dependent on the credit given to them by any association, but to the persuasion of the solidity of the Bank No association of individuals can prevent Bank notes from being depreciated compared with the standard if their quantity is excessive '

[98] Murphy, *John Law*, pp. 226, 229-30.

[99] *Monetary regimes in transition*, ed. Bordo and Capie, p 6 Henry Thornton's attitude, cited above, was less sanguine on the response of the public in the continental states which experienced paper money

[100] Bordo and White, 'British and French finance during the Napoleonic Wars', p 242 'Belief that ultimate budget balance would be restored meant that money creation, like borrowing, was a temporary measure.'

[101] *Monetary regimes in transition*, ed Bordo and Capie, p 3

[102] Hoffman, Postel-Vinay and Rosenthal, *Priceless markets*, pp. 109, 206, 275, ibid p 77 'If we apply our model to the notarial records, it becomes clear that few individuals perceived Law's System as stable.'

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At first, this gives government the short-term advantage in the rapid expansion of paper money: but it is an advantage which is rapidly eroded, as in France in the 1790s, once persistent inflation and depreciation of the paper become the expectation of the public.

The breakdown of the Bretton Woods system in 1971, once the dollar's link with gold was ended, has led to the acceptance that government fiat (in the sense of the law on legal tender) is the source of the value of money. Since then there have been limitless possibilities for expanding the money supply. In the first two decades of this new monetary regime, the world suffered rapid inflation and economic instability. One way of avoiding inflationary 'surprise' is to ensure that there is an independent monetary authority within a state whose main, if not only, function is to deliver stable prices. The first experiments in fiat money emanating from the French experience suggest that most European governments had little or no ability to prevent an inflation tax from being added to the burdens paid by their taxpayers, not least because the yield from direct and indirect taxation was insufficient for the needs of government in wartime. Conversely, in Britain, where the per capita tax burden was extremely heavy, Parliament acted as the custodian of the rights of property-owners to prevent simultaneously a repudiation of the debt and an additional wartime inflation tax caused by any excessive issues of the 'paper pound'

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